MultiFlow

Multimodal Interaction Toolkit

Group 2

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Project Overview

What is MultiFlow?

An interactive toolkit that fuses voice, gesture, and color inputs into a unified multimodal experience — built as a reusable library for any web application.

Voice

Web Speech API
Continuous Recognition
Command Dispatch

Gesture

MediaPipe Hands
EMA Smoothing
Coordinate Tracking

Color

Canvas RGB Sampling
Real-time Detection
Center Point Mapping

Architecture Overview

PRESENTATION LAYER

index.html • styles.css • script.js • Canvas Rendering Engine

ORCHESTRATION LAYER — MultiflowFusionEngine

State Manager • Event Router • Late Fusion Logic • Temporal Window (3s)

INPUT MODULES

VoiceModule | GestureModule | ColorDetectionModule

Modular Monolith • Package-by-Feature • JavaScript + HTML • 74.5% JS / 25.5% HTML

Monorepo: packages/multiflow-toolkit • apps/smart-paint

Multimodal Fusion Strategy

Late Fusion

Modalities processed independently, then combined at decision level

3 sec

Temporal fusion window

How Fusion Works

- ① Speech recognizer fires → command stored in temporal buffer
- ② Gesture module streams hand coordinates → decoded as drawing intent
- ③ Color module samples frame center → maps to closest named color
- ④ FusionEngine correlates events within 3-second window
- ⑤ Composite command dispatched e.g. "paint blue" → activates color + painting state

Demo App — Smart Paint

What Smart Paint Demonstrates

- Say "paint" → enables freehand drawing via hand tracking
- Say "background" + show color → sets canvas background
- Say "stop" → pauses drawing mode
- Say "clear" → wipes canvas
- Gesture coordinates → real-time line drawing on canvas
- Camera feed → color detection at frame center

Tech Stack

Web Speech API
Voice commands

MediaPipe Hands
Hand landmark tracking

Canvas 2D API
Drawing engine

EMA Smoothing
Gesture stability

RGB Color Mapping
Color detection

multiflow-toolkit — Core API

VoiceModule

```
start() / stop()
onCommand(cb)
setCommands(list)
```

GestureModule

```
start() / stop()
onGestureData(cb)
configure(params)
```

ColorDetection Module

```
start() / stop()
onColorUpdate(cb)
setMappings(map)
```

Multiflow FusionEngine

```
register(module)
setFusionRule(fn)
dispatch(event)
```

Implementation Status

✓ VoiceModule
✓ GestureModule (MediaPipe)
✓ ColorDetectionModule
✓ MultiflowFusionEngine (Late Fusion)
✓ Smart Paint demo application
○ Plugin/registry system for custom modalities

Improving Modularity & Generality

1. Plugin Registry Pattern

Replace hardcoded module imports with a register(module) API. Each module declares its own event types and capabilities, so new modalities (eye tracking, touchpad, EEG…) can be added without touching core code.

2. Abstract Base Class / Interface

Define a IModalityModule interface: start(), stop(), onData(cb), getCapabilities(). All three current modules should extend it — enabling generic fusion logic to work with any future module.

3. Configurable Fusion Rules

Expose a setFusionRule(fn) hook so apps can define custom fusion logic (e.g. early fusion, weighted voting) instead of being locked into the current Late Fusion + 3s temporal window.

4. TypeScript & Event-Driven Bus

Migrate to TypeScript for type safety on module events. Replace direct callbacks with a typed EventBus — decouples modules and enables replay/testing of event streams.

Next Application Idea: AccessFlow

Hands-Free Web Browser & Page Reader

A fully accessible browser interface controlled entirely by voice + gestures, demonstrating the toolkit's generality beyond creative apps.

Voice Control

- Say 'scroll down / up'
- Say 'click [link name]'
- Say 'go back / forward'
- Say 'read page' → TTS

Gesture Navigation

- Swipe left/right → tab switch
- Point at element → hover
- Pinch → zoom in/out
- Thumbs up → scroll down

Fusion Interactions

- Voice + point → 'click that'
- Color detection → highlight mode
- Eye dwell + voice confirm
- Temporal fusion: 3-step nav flows

Roadmap — What's Next

Now — Done

- VoiceModule
- GestureModule
- ColorDetectionModule
- FusionEngine (Late Fusion)
- Smart Paint demo

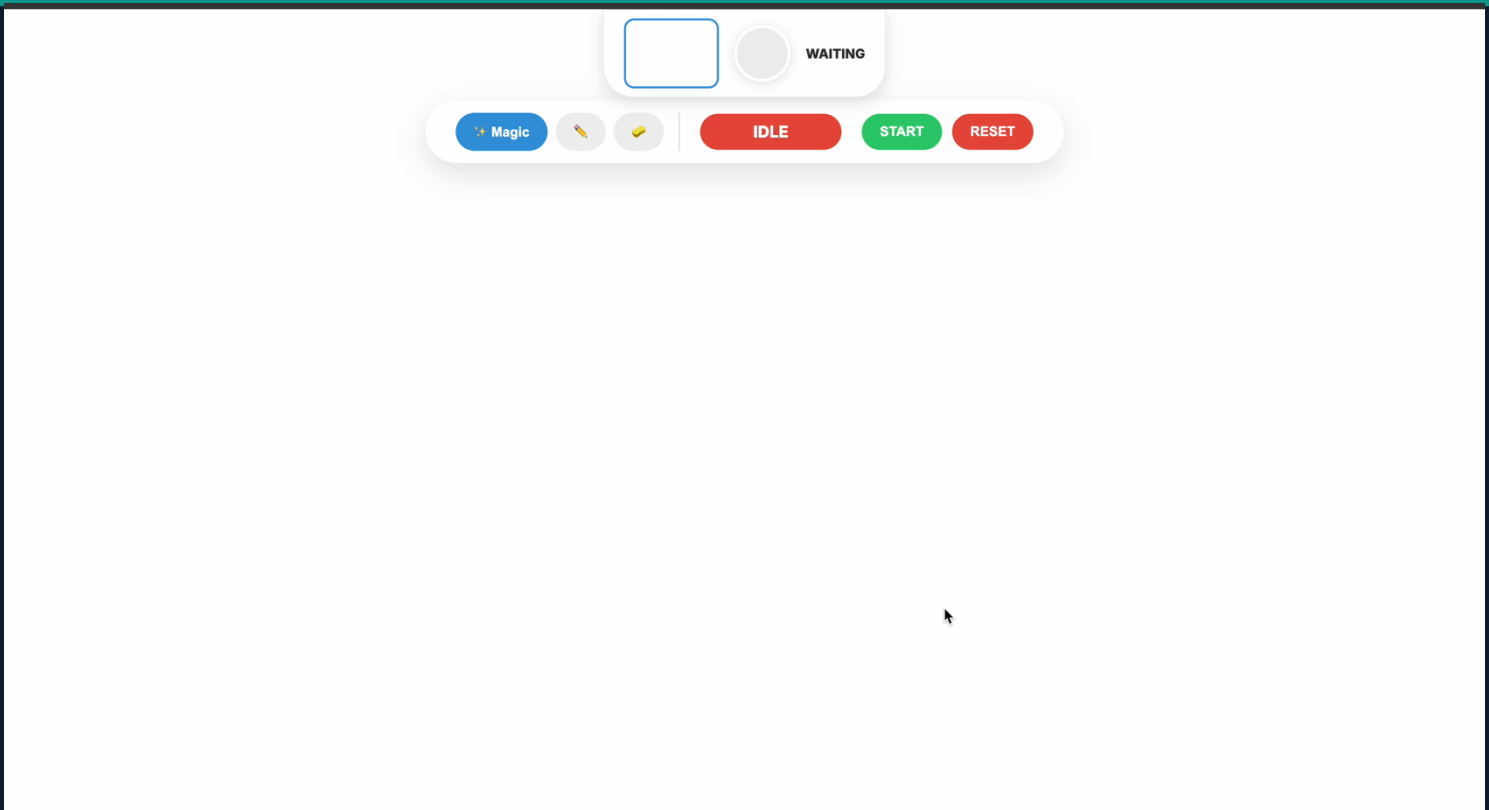
Phase 2 — Modularity

- IModalityModule interface
- Plugin registry
- Configurable fusion rules
- TypeScript migration

Phase 3 — AccessFlow App

- Hands-free browser UI
- Voice + gesture page nav
- TTS read-aloud
- Accessibility testing

Demo



Thank You!

Questions?